The empirical recurrence for OEIS A234242: recovering a conjecture that is not written down

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Abstract

OEIS A234242 records "Empirical recurrence of order 45", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 512$ vertices and satisfies some monic recurrence of order at most S; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 45, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 45 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A234242 is "Number of (n+1) X (2+1) 0..7 arrays with every 2 X 2 subblock having its diagonal sum differing from its antidiagonal sum by 2 (constant-stress 1 X 1 tilings).". It has offset 1 and begins

$$6706, 99630, 1520024, 23428586, 361889580, 5604275160, 86776363660, 1346057282610, \ \ldots$$

The entry states:

> Empirical recurrence of order 45 (see link above).

As of the "Last modified" line on the live entry (Jun 20 2022 19:09:47, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 512$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S =$ 512.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1024$ exact terms of the model returns order 45 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{45} c_i\, a(n-i),$$

c_1	46	c_{13}	−1075341235831351	c_{25}	−49698971791013793780127	c_{37}	6731574510
c_2	134	c_{14}	−9712651178468259	c_{26}	141332682505381310893989	c_{38}	2023442562
c_3	−35276	c_{15}	64089609807225875	c_{27}	183123165803850090714818	c_{39}	−362201889
c_4	323233	c_{16}	235935953639737470	c_{28}	−780626390348928587846808	c_{40}	−249427028
c_5	9930899	c_{17}	−2293634297261234632	c_{29}	−270395567855776777543560	c_{41}	1051812294
c_6	−157313817	c_{18}	−2725332118925920409	c_{30}	2714801171351407674882096	c_{42}	−291833556
c_7	−1188548411	c_{19}	52743969089997044624	c_{31}	−595246179658172059475072	c_{43}	−154724729
c_8	31938720142	c_{20}	−13241802902862635215	c_{32}	−5866168797635701662698496	c_{44}	681352545
c_9	27717326256	c_{21}	−796374230173916497493	c_{33}	3500666468919287746553856	c_{45}	908470060
c_{10}	−3554975670842	c_{22}	971308268134472771020	c_{34}	7599389065184082704108544		
c_{11}	8435049174150	c_{23}	7880760194879554361386	c_{35}	−6789152380252503977717760		
c_{12}	237054281056177	c_{24}	−15782086324909378071518	c_{36}	−5547706382650765698514944		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 45, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $45 + 4$ terms removed returns order 45 as well, so $\deg q_{\min} = 45 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $45 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 45 that this sequence satisfies — and that family turns out to have exactly one member.

References

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